

BR-32 Internal Measurement Rules and Tolerances

Measurement rules apply only when all required source inputs are explicitly present. Missing source fields must not be inferred from other records.

Rule ID	Topic	Rule
MR-001	Deck patch concrete volume	$\text{repair_area_m2} \times \text{average_repair_depth_m}$
MR-002	Approach slab volume	$\text{length_m} \times \text{width_m} \times \text{thickness_m}$
MR-003	Parapet curb volume	$\text{length_m} \times \text{width_m} \times \text{height_m}$
MR-004	Joint seal length	measured joint run length
MR-005	Rebar or dowel weight	$\text{bar_length_m} \times \text{unit_mass_kg_per_m}$
MR-006	Anchor grout volume	$\text{hole_count} \times \text{hole_volume_m3}$
MR-007	Overlay area	$\text{length_m} \times \text{width_m}$
MR-008	Crossfall percent	$\text{elevation_drop_m} / \text{lane_width_m} \times 100$
MR-009	Elevation variance	asbuilt_elevation_m minus design_elevation_m for the same explicit survey point

Tolerance ID	Check	Limit
TL-001	Deck elevation variance	absolute variance ≤ 0.010 m
TL-002	Crossfall	crossfall percent ≤ 2.50 percent
TL-003	Concrete ticket versus calculated volume	absolute difference ≤ 0.100 m3
TL-004	Joint seal length variance	absolute difference ≤ 0.100 m
TL-005	Concrete compressive strength	result ≥ 35.0 MPa
TL-006	Concrete air content	$4.0 \text{ percent} \leq \text{result} \leq 7.0 \text{ percent}$
TL-007	Anchor pullout	result ≥ 55.0 kN
TL-008	Core thickness	result ≥ 75.0 mm
TL-009	Material certificate reference	certificate reference must be nonblank

Status	Definition
SOURCE_OK	The value is directly supported and no conflicting source record is found.
CONFLICT	Another supplied file gives a different value for the same span, lane, repair zone, station range, pay item, measurement type, and unit.
MISSING	A required source field is blank or absent.
DUPLICATE	The same source repeats the same span, lane, repair zone, station range, pay item, measurement type, unit, and value.
TEXT_ONLY	The record is descriptive and has no numeric quantity.